

Module 08: Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form (BCNF) Architecture

Strict Determinant Requirement: Every LHS Must Be a Super Key & 3NF vs BCNF Trade-Offs

Boyce-Codd Normal Form (BCNF / Strict 3.5NF):

Relation R is in BCNF iff for every non-trivial functional dependency $X \rightarrow Y$:

X MUST BE A SUPER KEY OF R. (No exceptions! Zero prime relaxers!)

3NF vs BCNF Strict Comparison

Mathematical Inclusions & Trade-Offs

Inclusion Hierarchy

$BCNF \subset 3NF \subset 2NF \subset 1NF$

Every BCNF is 3NF. But not every 3NF is BCNF!

Lossless Join Property

- 3NF Decomposition: Always Lossless (Non-additive).
- BCNF Decomposition: Always Lossless.

Dependency Preservation (The Critical Divergence)

- 3NF: ALWAYS preserves all functional dependencies.
- BCNF: MAY LOSE functional dependencies upon split!

Classic Gateway BCNF Problem

Relation R(Student, Subject, Teacher)

Given Dependencies:

1. (Student, Subject) $\rightarrow$ Teacher
2. Teacher $\rightarrow$ Subject (Each teacher teaches 1 subject)

Candidate Keys: {Student, Subject} and {Student, Teacher}

Why it is in 3NF but NOT in BCNF:

In Teacher $\rightarrow$ Subject:

- Teacher is NOT a Super Key (Fails BCNF!).
- Subject IS a Prime Attribute (Passes 3NF!).

Decomposing into R1(Teacher, Subject) & R2(Student, Teacher)

LOSES the dependency (Student, Subject) $\rightarrow$ Teacher!

> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Boyce-Codd Normal Form (BCNF), Strict Super Key Determinant Condition, BCNF vs 3NF Deep Comparison, The Classic Database Trade-Off: Dependency Preservation vs Redundancy, Solved Gateway AKTU Numerical.

1. Motivation Behind BCNF

1974 me Raymond Boyce aur Edgar Codd ne observe kiya ki 3NF kuch specific conditions me redundancy aur anomalies ko poori tarah eliminate nahi kar pata.

Yeh situation tab aati hai jab ek table me **multiple overlapping Candidate Keys** hoti hain!
3NF ka relaxer rule ("Y\$ is a Prime Attribute") un dependencies ko allow kar deta hai jisme determinant Super Key nahi hota.

Is loophole ko close karne ke liye **Boyce-Codd Normal Form (BCNF)** introduce kiya gaya, jise aksar **3.5 Normal Form** bhi kaha jaata hai.

2. Formal Definition of BCNF

Ek relation schema \$R\$ **Boyce-Codd Normal Form (BCNF)** me tab hota hai jab:

1. Wo relation already **1NF** me ho.
2. Har non-trivial functional dependency \$X

ightarrow Y\$ ke liye:

\$X\$ **MUST BE A SUPER KEY OF \$R\$**

Yahan koi "OR" condition nahi hai! Determinant XY ko compulsory roop se Super Key hona hi padega!

3. Comprehensive Comparison: 3NF vs BCNF

Parameter	Third Normal Form (3NF)	Boyce-Codd Normal Form (BCNF)
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$Y \rightarrow X$ | XY is Super Key OR Y is Prime Attribute | XY MUST be Super Key |

Strictness	Relaxed (allows prime attribute on RHS)	Extremely strict (zero exceptions)
Lossless Join Decomposition	Always guaranteed (✓)	Always guaranteed (✓)
Dependency Preservation	Always guaranteed (✓)	NOT always possible (✗)
Storage Redundancy	Minimal (slight redundancy may remain)	Zero redundancy for functional dependencies
Practical Industry Choice	Most widely used in transactional DBs	Used when dependency loss is acceptable

4. Architectural Diagram

5. Classic Gateway Classes Solved Numerical: The Trade-Off Case

Problem: Given relation $R(\text{Student}, \text{Subject}, \text{Teacher})$ with functional dependencies:

$$F = \{ \text{Student} \rightarrow \text{Subject} \}$$

$$\text{Teacher} \rightarrow \text{Teacher}$$

$$\text{Subject} \rightarrow \text{Subject}$$

(Assumption: Har student ek subject ke liye ek teacher ko assign hota hai, aur har teacher sirf ek subject padhata hai).*

Step 1: Candidate Keys

$(\text{Student}, \text{Subject})^+ = \{ \text{Student}, \text{Subject}, \text{Teacher} \} = R \implies (\text{Student}, \text{Subject})$ is Candidate Key 1.

$(\text{Student}, \text{Teacher})^+ = \{ \text{Student}, \text{Teacher}, \text{Subject} \} = R \implies (\text{Student}, \text{Teacher})$ is Candidate Key 2.

Prime Attributes: $\{ \text{Student}, \text{Subject}, \text{Teacher} \}$ (All attributes are Prime!).

Step 2: 3NF Test

1. $(\text{Student}, \text{Subject})$

$$\text{Teacher} \rightarrow \text{Teacher}$$

- LHS $(\text{Student}, \text{Subject})$ is a Candidate Key (Super Key). $\implies$ Satisfies 3NF.

2. Teacher

$\rightarrow \text{ext}\{\text{Subject}\}$:

- LHS $\text{ext}\{\text{Teacher}\}$ is NOT a Super Key.
- RHS $\text{ext}\{\text{Subject}\}$ IS a Prime Attribute (part of Candidate Key 1).
- $\implies$ **Satisfies 3NF!**

Verdict for 3NF: Relation R is in **3NF!**

Step 3: BCNF Test

Look at $\text{ext}\{\text{Teacher}\}$

$\rightarrow \text{ext}\{\text{Subject}\}$:

- Determinant $\text{ext}\{\text{Teacher}\}$ is **NOT a Super Key!**
- In BCNF, there is NO prime attribute relaxation!

Verdict for BCNF: Relation R **FAILS BCNF!**

Step 4: Decomposing into BCNF & The Great Loss

To bring into BCNF, decompose using the violating dependency $\text{ext}\{\text{Teacher}\}$

$\rightarrow \text{ext}\{\text{Subject}\}$:

$R_1(\text{Teacher}, \text{Subject})$ with Key = $\text{ext}\{\text{Teacher}\}$ (in BCNF).

$R_2(\text{Student}, \text{Teacher})$ with Key = $\{\text{ext}\{\text{Student}\}, \text{ext}\{\text{Teacher}\}\}$ (in BCNF).

Both sub-relations are in BCNF and the join is Lossless ($R_1 \cap R_2 = \{\text{ext}\{\text{Teacher}\}\}$, which is a key in R_1).

The Critical Flaw: Original dependency $\{\text{ext}\{\text{Student}\}, \text{ext}\{\text{Subject}\}\}$

$\rightarrow \text{ext}\{\text{Teacher}\}$ cannot be tested in either R_1 or R_2 alone!

Dependency Preservation is LOST!